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MILITARY SPECIFICATION

# LATHES, ENGINE AND TOOL ROOM 13 THRU 17 INCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE .

1.1 Scope. This specification covers electric motor driven, metal turning and threading, engine and toolroom lathes. The lathes shall have either geared head drive or combination gear and belt drive with equipment and accessories as described herein.

1.2 Classification. This specification covers lathes of the following types and sizes. The type and sizes furnished shall be as specified, (see 6.2.1).

Type I - Engine lathes

Type II - Toolroom lathes

| Size | Swing over bed | | Swing over cross slide | |
|---|---|---|---|---|
| | Inches | (mm) | Inches | (mm) |
| 1307 | 13 | (330) | 7 | (177) |
| 1508 | 15 | (381) | 8 | (203) |
| 1610 | 16 | (406) | 10 | (254) |
| 1709 | 17 | (431) | 9 | (228) |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3416

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## 2. APPLICABLE DOCUMENTS

### 2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification-to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. So DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE

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ANSI/B5.l - T-slots, their Bolts, Nuts, and Tongues.
ANSI/B5.8 - Chucks and Chucks Jaws.
ANSI/B5.9 - Spindle Noses for Toolroom Lathes, Engine Lathes, Turret Lathes, and Automatic Lathes (IS0 702).
ANSI/B5.l0 - Machine Tapers.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018. )

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Commercial Packaging, Standard Practice for. .
ASTM-E-380 -Metric Practice Guide.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 79 - ANSI C113.1 Electrical Standard for Metalworking Machine Tool and Plastics Machinery.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and reference cited herein,. the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, (see 6.2.1) a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.2 Design. The lathe shall be new and one of the manufacturer' s current models capable of conforming to the accuracy requirements for an engine lathe, or tool room lathe as specified herein. The lathe shall be capable of performing turning, boring, facing and threading operations within the capacities of the size specified. The lathe shall include all components, parts, and features necesary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.



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3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) system of measurement is acceptable. The particular system required shall be as specified (see 6.2.1). If Customary System of Units (US) is specified, means shall be provided to permit conversions to SI units. In this specification, all measurements, dimensions, sizes and capacities are given in the US. These measurements may be converted to the International System of Units (SI) through the use of the conversion factors and methods specified in ASTM-E-380. When specified (see 6.2.1), a digital readout system providing conversion characteristics outlined above, shall be furnished.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system..

3.2.1 .1.1 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI system of measurements. When a dual US and SI system is furnished, both feed dials shall have independent zero adjustments and both dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine’s intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The lathe and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.



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3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall-not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The machine shall be designed and constructed such that, under the operating, service, transportation and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by federal, state or local statutes in effect at the point of the installation.

3.2.9 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating Lubrication’systems shall include a filter which is cleanable or replaceable. Each lubricant l reservoir shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible.

3.2.10 Interchangeability. To provide for replacement of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification to the part or the machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the



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machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer’s commercial practice.

3.3.5 Painting. All surfaces to be painted shall be cleaned of all foreign matter. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with manufacturer's standard commercial practice which shall be not less than one coat of primer and one coat of finish color.

3.3.6 Threads. Except for the leadscrew, when applicable, all threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All dials that indicate stock removal or tool movement shall be graduated in increments not larger than 0.001 inch. Dial diameters shall be such that graduations may be read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a nonglare background.

3.3.8 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.3.9 Gears. All gears shall be constructed from a material suitable for the intended purpose. All metal gears shall remachined and heat treated by a process that will impart the hardness, surface finish and toughness that will enable the gear train to transmit full-rated torque of the drive motor without gear damage, failure or premature deterioration and wear. All gears shall be manufactured to definite established standards to provide for interchangeability and parts replacement.

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3.3.10 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with the requirements of MIL-STD-461,

3.4 Components. Each lathe shall consist essentially of a bed and base, headstock, headstock spindle, spindle drive, spindle speed change mechanism, carriage and apron, feed mechanism, cross slide, compound rests carriage feed and tailstock. In addition, the machine shall have all other equipment specifically described herein and all additional equipment necessary for the machine to meet all of the performance requirements stated herein.

3.4.1 Bed and base. The lathe shall have a bed and base structure that supports the headstock, tailstock, carriage, and other related equipment. The-structure shall have sufficient strength and rigidity to support all the components in rigid alignment in accordance with table II. The bed shall have V-ways or flatways or a combination of “V” and flat ways to provide a sliding surface for the longitudinal movement of the carriage and assure alignment of the headstock and tailstock. The bed ways for the carriage longitudinal movement shall be hardened and ground to minimize wear and maintain accuracy of alignment. The bed ways shall be either cast integrally with the bed structure or may be replaceable way inserts, or a combination thereof. Means shall be provided to lubricate the bedways automatically or by one-shot lubrication. The bed shall have a sufficient number of covered access openings to facilitate inspection, adjustment and repair of components located within the structure. Means for leveling the base of the machine shall be provided.

3.4.2 Chip and coolant pan. The chip and coolant pan shall be constructed to allow coolant to drain from the chips and be returned to the coolant system for reuse. There shall be adequate clearance between the chip pan and the lathe bed to allow for chip removal.

3.4.3 Headstock. The lathe shall have a headstock that supports and aligns the spindle axis parallel with the bed ways. The headstock shall house the gearing for the spindle drive and power take-off source for the feed mechanism. The headstock shall be constructed to support the spindle and gear shafts so as to prevent deflection that would affect accuracy under full operating and cutting loads. The headstock shall be an oil tight unit and shall have means of lubricating all bearings and gears that require continuous lubrication. The headstock shall include necessary controls for selecting and changing spindle speeds.

3.4.4 Headstock spindle. The lathe shall have a headstock spindle to support a chuck, faceplate, or collet which in turn holds and provides rotation of the workpiece. The spindle shall be supported by roller or ball type bearings that have a size, type and precision class suitable for the application. Unless otherwise specified (see 6.2.1), the spindle nose shall be the type “D” as normally provided as specified in table I and shall conform to the requirements of ANSI B5.9. The spindle internal taper shall be as specified in table I and shall conform to the requirements of ANSI B5.1O. The spindle nose shall be accurately machined to provide locating surfaces

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for lathe centers, faceplates and chucks. The inner and outer surfaces of the spindle nose on which work holding devices are mounted shall be hardened to prevent wear.

3.4.5 Spindle drive. The lathe shall have a spindle drive mechanism that will provide the required range of spindle speeds listed in table I. The spindle shall be driven through an all gear or combination gear and belt drive by an electric motor as specified in table I. All gears shall meet the requirements of 3.3.9.

3.4.6 Spindle speed change mechanism. The spindle speed mechanism shall be the manual speed change, the manual speed change with pre-select power shift or the variable speed type. The shifting mechanism shall be designed so that speed change cannot occur unintentionally. Except for the variable speed drive, speeds shall be in geometric progression within 10%. The manual speed change mechanism shall allow spindle speed changes by positioning gear shift levers or dials located on the front side of the headstock. Speeds shall be selected by positioning the levers as indicated on a conveniently located speed chart or by positioning a direct reading dial. The manual speed change with pre-select power shift shall provide spindle speed selections by manually-setting a selector control to the desired spindle speed. It shall be possible to pre-select the next spindle speed while still cutting, then automatically initiate the shift upon activation of the spindle control. The variable speed change mechanism shall permit spindle speed adjustments to compensate for varying machine conditions. Optimum speeds shall be obtained by gear range selection and by variable speed adjustment within the range. It shall be possible to adjust speed within a gear range while running. The actual spindle speed in RPM shall be shown by a speed indicating device located convenient to the operator.

3.4.7 Feed-thread gearbox. The lathe shall have a feed-thread gearbox which changes the ratio of the speed of the spindle with respect to the lead screw and feed rod to obtain the-feed rate and thread lead required. Unless otherwise specified (see 6.2.1), the lathe shall be capable of producing inch and metric threads over the thread ranges specified in table I. The feeds shall be by a geared feed mechanism providing both manual and power operation. Power to the feed mechanism shall be through a separate feed rod and lead screw. A safety clutch or similar device shall protect the feed mechanism from damage by overload or jamming. It shall be possible to feed and thread in either direction independently of the direction of spindle rotation and to provide a means to reverse the leadscrew when cutting metric threads. A feed and thread chart shall be mounted near the gearbox as a ready reference for feed and thread selections. All feed and thread selections shall be shown without need for mathematical conversion. When specified (see 6.2.1), a combination inch/metric gearbox shall be furnished.

3.4.8 Carriage. The lathe shall have a carriage that supports the apron, cross slide, compound rest, and other related equipment. The carriage shall traverse longitudinally on the bed ways. The carriage shall be fitted to the bed ways and shall have means of compensating for wear between sliding



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surfaces. Means shall be provided to lock the carriage to the bed at any desired location between the headstock and tailstock for cross feed type operations. The carriage shall operate in both longitudinal directions under power, regardless of the direction of spindle rotation. The carriage wings shall have means to prevent chips and abrasive materials from entering between carriage slide and bed ways. Provision shall be made for use of taper attachment and follow rest. The top of the carriage shall have ways perpendicular to the axis of the machine to support the cross slide and compound rest.

3.4.9 Apron. The lathe shall have an apron on the front side of the carriage which shall house the geared feed mechanism that provides the powered longitudinal, cross, and thread chasing feeds. The apron shall be of double wall construction, providing bearing support at both ends of gear shafts. The apron shall support the handwheel that provides manual longitudinal feeds. The lathe shall have a separate feed rod and lead screw. A safety interlock shall prevent engaging the feed control and threading control simultaneously. A clutch or similar device shall protect the feed mechanism against damage from overload. A thread dial indicator shall be mounted on the apron to indicate the position to engage the half-nut with the lead screw when cutting threads. Automatic lubrication shall be provided for all moving parts within the apron. Automatic or manually operated one shot pressure lubrication shall be provided for way surfaces.

3.4.9.1 Apron controlled leadscrew reverse. When specified (see 6.2.1), the lathe shall have an apron controlled leadscrew reversing mechanism to permit reversing the direction of the carriage without disengaging the leadscrew half-nut. When specified (see 6.2.1), feed stops shall automatically disengage carriage feeds when turning or threading in either longitudinal direction of travel.

3.4.10 Cross slide. The cross slide shall be fitted to the carriage ways and shall have means to compensate for wear between the sliding surfaces. The cross slide shall be positioned by a precision feed screw which shall be protected from dirt and chips.

3.4.11 Compound rest. A compound rest shall be mounted on top of the cross slide. The base of the compound rest shall be graduated in one degree increments to not less than 90 each side of a zero reference. The base shall swivel horizontally through 360 degrees. Means shall be provided to lock the compound at any angular setting. The compound rest shall have a standard size T-slot for mounting tool holders. The compound slide feed screw shall be protected from dirt and chips.

3.4.12 Tailstock. The tailstock shall be constructed to permit supporting a work part close to the compound rest without interference with operation of the top slide. The tailstock shall be easily moved along the bedways and have a quick acting clamp for locking in position. The spindle shall hold and eject tang type tools, and shall be graduated in increments



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not greater than 1/16 inch for measuring spindle extension. The tailstock spindle taper shall be in accordance with ANSI B5.10 for self-holding tapers. The tailstock shall be marked to indicate set-over as specified in table I. Spindle locking shall not affect accuracy of alignment.

3.4.13 Coolant system. The coolant system shall include a sump or reservoir, a filter, a power driven pump and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

3.4.14 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA-79. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 230 volts. A wiring diagram shall be mounted near the input power junction providing wiring connections for 230 volt and 460 volt operation. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA-79 for the disconnect fuse rating.

3.4.15 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure.

3.5. Size and capacities. Unless otherwise specified (see 6.2.1), the size and capacities of the machine shall be not less than those specified in table I for the machine size specified.

3.6 Alignment tolerances. Each machine shall meet the alignment accuracies of table 11 when tested in accordance with 4.5.2.

3.7. Performance. The machine shall perform to the accuracies of table III, when tested in accordance with 4.5.3.

3.8 Standard equipment. Unless otherwise specified one set of hand tools, normally furnished by the manufacturer, shall be furnished. Additional equipment required shall be furnished as specified and fully described (see 6.2.1).

3.8.1 Lathe centers and centers sleeve. One headstock center, one tailstock center and one headstock center sleeve shall be furnished. The center rests and sleeves shall conform to the requirements of ANSI B5.9 for the type spindle furnished.



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TABLE I - Sizes and capacities.
Type I and Type II

| Characteristics | Size - Dimensions in inches | | | |
|---|---|---|---|---|
| | 1307 | 1508 | 1610 | 1709 |
| Swing over bed, inches | 13 | 15 | 16 | 17 |
| Swing over cross slide, inches | 7 | 8 | 10 | 9 |
| Distance between centers, inches: | | | | |
| Standard | 25 | 30 | 40 | 40 |
| Optional | 40,50,60 | 42,50,54 | 54,60,78 | 30,54,60,78,100 |
| Spindle speed selections | | | | |
| Standard | 12 | 12 | 12 | 12 |
| Optional | - | 16 | 16 | 16 |
| Spindle speed ranges, RPM: | | | | |
| Standard | 50-1800 | 50-1800 | 40-1600 | 40-1250 |
| Optional | 50-220 | 30-1200 | | 25-2000 |
| Cross slide travel, inches (W/O taper) | 7-3/8 | 8-1/2 | 8-1/2 | 10 |
| Compound slide travel, inches | 3-1/2 | 3-3/4 | 5 | 5 |
| Main drive motor, HP: (Continuous) | | | | |
| Standard | 3 | 5 | 7-1/2 | 7-1/2 |
| Optional | 5 | 7-1/2, 10 | 10 | 10 |
| Size hole through spindle, inches | 1-9/16 | 2 | 2 | 3 |
| Spindle nose size, type | 4"D-1 | 6"D-1 | 6"D-1 | 8"D-1 |
| Spindle center, Morse taper | #3 | #3 | #4 | #4 |
| Longitudinal feed selections | 20 | 45 | 28 | 30 |
| Longitudinal feeds, IPR | .0018-.0287 | .0025-.040 | .0025-.028 | .0025-.032 |
| Threads, English: | | | | |
| Number of selections | 35 | 45 | 35 | 35 |
| Number of threads, TPI | 2 to 56 | 2 to 64 | 2 to 64 | 2 to 56 |
| Threads, Metric: | | | | |
| Number of selections | 35 | 35 | 35 | 40 |
| Number of threads, mm | .4 to 7 | .25 to 10 | .4 to 14 | .5 to 12 |
| Tailstock spindle diameter, inches | 1-5/8 | 1-7/8 | 2-7/8 | 2-7/8 |
| Tailstock spindle travel, inches | 4-1/2 | 5 | 6 | 6 |
| Tailstock center, Morse taper | #3 | #3 | #4 | #4 |
| Tailstock set over, each side of center inches | 3/8 | 3/8 | 3/8 | 3/8 |
| Leadscrew diameter, inches | 1-1/8 | 1-3/16 | 1-1/4 | 1-1/4 |

NOTE: The machine size and capacity shall not be leas than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above table. When the largest size and capacity of the machine offered shall not exceed the Stated requirements by more than 10 percent.



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TABLE II - Alignment accuracies.

| Test | Type I Engine lathe | Type II Toolroom lathe |
|---|---|---|
| Spindle center runout | 0.0005 in. TIR | 0.0004 in. TIR |
| Spindle nose runout | 0.0004 in. TIR | 0.0003 in. TIR |
| Cam action of spindle, rear side of test plate | 0.0005 in. TIR | 0.0003 in. TIR |
| Spindle taper runout: | | |
| at end of 12 in. test bar | 0.0008 in. TIR | 0.0006 in. TIR |
| at spindle nose end of bar | 0.0004 in. TIR | 0.0003 in. TIR |
| Headstock alignment: | | |
| Vertical, high at end of 12 in. test bar | 0.001 in. | 0.0005 in. |
| Horizontal, at end of 12 in. test bar toward operators side | ±0.0005 in. | ±0.0003 in. |
| Headstock and tailstock centers alignment, vertical, high at tailstock | 0.003 in. | 0.002 in. |
| Leadscrew, lead per foot | ±.0015 in. | ±0.001 in. |
| Cross slide alignment, to face | | |
| Hollow or concave only, on 12 in. diameter | 0.001 in. | 0.0005 in. |
| Face plate runout: | | |
| on diameter | 0.001 in. | 0.0005 in. |
| on face at nominal diameter | 0.0015 in. | 0.001 in. |
| Chuck-runout, face and periphery | 0.003 in. | 0.003 in. |

TABLE III - Performance accuracies.

| | Engine Type I | Toolroom Type II |
|---|---|---|
| Rough turning: | | |
| Out-of-round | 0.005 inch | 0.005 inch |
| Taper per foot | 0.005 inch | 0.005 inch |
| Finish turning: | | |
| Out-of-round | 0.0004 inch | 0.0003 inch |
| Cylindrical - Two finished collars shall have the same diameter within (see 4.5.3.2) | | |
| Part held in chuck or collet | 0.0015 inch | 0.0008 inch |
| Part held between centers | 0.0008 inch | 0.0004 inch |



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3.8.2 Chuck and chuck jaws. The number and type of chuck and chuck jaws shall be as specified (see 6.2.1), and shall conform to the applicable requirements of ANSI 65.8.

3.8.3 Drive plate. The drive plate shall be the maximum allowable diameter, not to exceed the nominal swing over the bedways. The plate shall have not less than two radial slots and shall have at least one through slot for driving lathe dogs.

3.8.4 Taper attachment. The taper attachment shall be for mounting on the rear of the carriage. It shall be graduated for settings in both degrees and inches of taper per foot. Means for maintaining adjustment shall be provided. The attachment shall turn either right of left hand tapers. The taper attachment shall not interfere with the normal operation of the lathe.

3.8.5 Thread-chasing dial. The thread-chasing dial shall be calibrated to indicate correct engagement of the lead-screw and half-nut when cutting threads. A threading-chasing stop shall be provided.

3.8.6 Micrometer carriage stop. The micrometer carriage stop shall be mounted on the front bed way to provide an adjustable stop. The micrometer dial shall be graduated in increments of 0.001 inch or less with at least .750 inch of adjustment.

3.8.7 Work light. The work light shall be an adjustable arm type with all necessary components for mounting on the machine tool. The light shall operate on 110 to 120 volts alternating current, 60 hertz, and shall be shielded against dirt and chips.

3.8.8 Wrenches. All special tools and wrenches required for operating the lathe and accessories shall be provided.

3.9 Optional equipment. Optional equipment shall be furnished as specified and as fully described (see 6.2.1).

3.10 Additional requirements. Performance features, attachments or accessories required in addition to, or in lieu of those specified herein, shall be as required by the procuring activity (see 6.2.1).

3.11 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with the requirements of MIL-STD-461, Class _____, Requirement _____, (see 6.2.1).

3.12 Marking of instruction plates, indicating plates and namplates. All words on instruction plates, indicating plates and name plates should be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background. When these methods are not practicable, the instructions shall be applied directly on the item by environmentally protected decalcomania transfer, metal-wrap-around tag, stencil, silk screen or by any other method suitable for the use intended which shall be as permanent as the normal life expectancy of the item to which it is applied and capable of withstanding the environmental tests and cleaning procedures specified for the item.

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3.12.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.12.2 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion-resistant metal nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard lathe and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Type Size Horsepower
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.13 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract.

3.14 Workmanship Workmanship of the lathe and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article inspection is required, it shall be applled to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the item to pass the first article tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of at least the examination in 4.5 and the tests in 4.6.1. Failure of the item to pass the examination or tests shall be cause for rejection.

4.5 Examination. The machine and equipment shall be examined to determine compliance with the requirements of 3.2 through 3.7.

4.6 Tests.

4.6.1 Operational test. The lathe shall be operated at no load for not less than 60 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.6.2 Alignment accuracy tests. Each lathe shall be tested to determine that it complies with the alignment accuracy requirements of table II (see 3.6).

4.6.3 Performance tests. The lathe shall be subjected to the following turning tests to determine conformance to the accuracies of table III (see 3.7).

4.6.3.1 Round turning test. A bar of SAE 1045 carbon steel or steel of similar machinability, not less than 2 inches in diameter and 4 inches long, shall be held in the chuck furnished with the lathe. A rough and finish cut shall be made at least 2 inches long using a cutting speed of not less than 80 feet per minute. The turned diameter shall conform to the accuracy of the round turning test in table III.

4.6.3.2 Cylindrical turning test. Using the same type bar as for the round turning test, the bar shall be not less than 4 inches in diameter and 12 inches long. The bar shall be semi-finished with a collar at each end of the bar. The collars shall be not less than 4 inches apart. A finish cut shall be taken over the two collars in one pass using a single point carbide tip turning tool and a cutting speed not less than 350 feet per minute. The



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turned diameters shall conform to the accuracy of table III Upon completion of the test, the test specimen shall be prepared for turning between centers. A single cut shall be taken over the two collars and the turned diameters shall conform to the accuracy of the cylindrical turning test in table 111.

4.6.3.3 Horsepower test. Each machine shall be subjected to a test which applies not less than the full rated input power to the spindle motor for not less than one minute. Spindle speed shall be chosen by the supplier at the lowest practicable for this test without applying torque in excess of the design maximum value to any machine components. Material and dimensions of the workpiece, tooling, depth of cut, and feed rate shall be at the option of the supplier, to provide the required loading. There shall be no apparent evidence of chatter, stalling, or permanent distortion during or after the test.

4.6.4 Electromagnetic interference control tests. Equipment requiring electromagnetic Interference control shall be tested for compliance with 3.11.

4.7 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging, packing and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM-D-3951. When specified (see 6.2.1), level A packaging, level A or level B packing and special marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The lathes covered by this specification are intended for use in any machine shop for production or toolroom use where the nature of the work does not require the capabilities of a heavy duty lathe. The lathes are suitable for turning, boring, facing, threading, and taper turning using high speed steel or carbide tooling.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class and size required, etc. (see 1.2).

c. When first article is required for inspection and approval (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required systems (see 3.2.1).

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e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Dual calibrations (US and SI), if required (see 3.2.1.2).

g. Additional safety and health requirements, if required (see 3.2.5 and 6.4).

h. Painting, if different (see 3.3.5).

i. Spindle nose, indifferent (see 3.4.4).

j. Specify inch only or metric only, if required (see 3.4.7).

k. Specify combination inch/metric gearbox, if required (see 3.4.7).

1. Specify apron controlled leadscrew reverse, if required (see 3.4.9.1).

m. Electrical system and input power supply, if different (see 3.4.14).

n. Motor enclosure, if different (see 3.4.15).

o. Size and capacity, if different (see 3.5).

P. Standard equipment, if different (see 3.8).

q. Specify chuck size and chuck jaws required (see 3.8.2).

r. Optional equipment, as required (see 3.9).

s. Additional requirements, as required (see 3.10).

t. Electromagnetic interference control, if different cite the applicable class and requirements, (see 3.11).

u. Lubrication chart or plate, if different (see 3.12.1).

v. Nameplate, if different (see 3.12.2).

w. First article inspection, indifferent (see 4.3).

x. Quality conformance inspection, if different (see 4.4).

y. Level of packaging, packing and marking, if different (see 5.1).



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6.2.2 Contract data requirements. All data requirements such as parts lists, wiring diagrams, operating manuals, maintenance manuals, alignment and test results shall be requested on a DD Form 1423, Contract Data Requirements List and shall be furnished in accordance with each specified DD Form 1664, Data Item Description, as shown on the DD Form 1423.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial machines supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of the workplaces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1.

6.5 Supersession data. This specification includes the requirements of engine and toolroom lathes, size 13" thru 17", which were formerly included under type I, class A and type 11, class A of MIL-L-23400C dated 6 March 1981.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes, with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army -AL
Navy - SH
Air Force - 99

Review Activities:
Air Force - 84
DLA - GS

User Activities:
Army -ME, SM
Navy -AS, MC, YD

Preparing Activity:
DLA - IP

Project Number:
3416-0189



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-L-23249D | Lathes, Engine and Tool Room 13 thru 17 Inch |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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